**MARIWASA SIAM CERAMICS INC.**
TALENT ACQUISITION & TECHNOLOGY OPERATIONS

ENTERPRISE STANDARD OPERATING PROCEDURE & END-USER MANUAL

# Automated Resume Analysis System

## An Enterprise-Grade AI Ecosystem for High-Volume Talent Screening, Natural Language Resume Parsing & Intelligent Match Scoring

| | |
|---|---|
| DOCUMENT REFERENCE<br>SOP-HR-ARAS-2026-v2.4 | SOFTWARE VERSION<br>Release 2.4.0 (Enterprise Production) |
| TARGET AUDIENCE<br>Candidates, HR Staff, Recruiters, Administrators | EFFECTIVE DATE<br>November 2026 |
| CLIENT FACILITY<br>Mariwasa Plant, Sto. Tomas, Batangas | SECURITY CLASSIFICATION<br>RESTRICTED / INTERNAL BUSINESS USE |

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Powered by Next.js 15, FastAPI, Celery, spaCy NLP & Gemini AI

SYSTEM GOVERNANCE & DOCUMENT CONTROL

# Document Information & Approval History

This document serves as the official operational manual and system administration specification for the **Automated Resume Analysis System (ARAS)** deployed at **Mariwasa Siam Ceramics Inc.** It outlines end-to-end workflows for Candidates, HR Screening Personnel, and System Administrators, detailing functional interfaces, algorithmic scoring formulas, voice interview integrations, and operational recovery protocols.

> **Document Scope & Distribution Notice**
> The policies, procedures, and technical configurations contained herein apply to all human resource operations, factory plant hiring cycles, and administrative governance for Mariwasa Siam Ceramics Inc. Reproduction or distribution outside authorized organizational channels requires written permission from the Chief Information Officer and the Head of Human Resources.

## Document Control Summary

| Attribute | Specification Details |
|---|---|
| **Document Title** | Automated Resume Analysis System — Enterprise User Manual & Operations Guide |
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## Revision & Change Management Log

| Version | Release Date | Primary Author | Summary of Structural Changes & Feature Enhancements |
|---|---|---|---|
| `V1.0.0` | May 2026 | Project Engineering | Initial prototype release. Deterministic regex-based resume parsing, basic PostgreSQL relational schemas, and manual candidate listing. |
| `V1.5.0` | July 2026 | NLP Core Team | Integration of spaCy PhraseMatcher with 2,000+ domain taxonomy. Introduced weighted matching scoring (Skills, Experience, Education). |
| `V2.0.0` | October 2026 | Fullstack Team | Architectural migration to Next.js 15 App Router and FastAPI Asynchronous Celery workers. Added Cloudflare Tunnel orchestration and Redis caching. |
| `V2.2.0` | Late Oct 2026 | Voice & ATS Team | Added Twilio + Gemini Real-Time Voice Interview Caller (Bilingual Filipino/English) and Physical Hard-Copy Resume Storage Vault with Cabinet/Drawer tracking. |
| `V2.4.0` | November 2026 | Release Engineering | Enterprise Production Edition. Dynamic runtime scoring weights, Google Calendar OAuth sync, multi-candidate comparison, and self-service Preview & Verify flow. |

## Formal Approvals & Sign-Off

| Stakeholder Role | Name & Title | Status | Verification Date |
|---|---|---|---|
| **Lead System Architect** | STI CS Thesis Engineering Lead | APPROVED | November 10, 2026 |
| **HR Operations Lead** | HR Screening Lead, Mariwasa | APPROVED | November 11, 2026 |
| **Talent Acquisition Head** | Director of HR & Admin, Mariwasa | APPROVED | November 12, 2026 |
| **Chief Technology Officer** | Academic & Industrial Supervisor | APPROVED | November 13, 2026 |

REFERENCE GUIDE

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**Quick Reading Recommendation**

- **Candidates / Applicants:** Focus directly on **Chapter 2 (Access)** and **Chapter 3 (Candidate Guide)** for step-by-step instructions on resume submission and application tracking.
- **HR Recruiters & Hiring Managers:** Proceed directly to **Chapter 4 (HR Portal)** for instructions on candidate ranking, side-by-side comparison, interview booking, and hard-copy archiving.
- **System Administrators & DevOps:** Review **Chapter 5 (Admin Control)**, **Chapter 6 (AI Engine)**, and **Chapter 7 (Deployment)**.

CHAPTER 1 • SYSTEM ARCHITECTURE

# Executive Summary & System Architecture

## 1.1 Enterprise Background & Operational Objectives

**Mariwasa Siam Ceramics Inc.**, a premier pioneer in ceramic tile manufacturing operating under the SCG Group in Sto. Tomas, Batangas, processes thousands of job applications annually across plant manufacturing, quality assurance, engineering maintenance, supply chain, and administrative departments. Prior to the introduction of the **Automated Resume Analysis System (ARAS)**, the human resources team experienced severe screening bottlenecks, subjective qualification discrepancies, high candidate interview no-show rates, and difficulties tracking hard-copy resumes submitted directly at plant security gates.

### Previous Operational Bottlenecks

- **Screening Latency:** Manual review of 1,000+ resumes required 3-4 weeks per hiring cycle.
- **Subjective Evaluations:** Inconsistent scoring across different recruiters and departments.
- **Hard-Copy Filing Chaos:** Physical walk-in resumes misplaced in physical storage drawers.
- **Candidate Interview Drop-Off:** Low confirmation rates via standard email notifications.

### ARAS System Transformations

- **Sub-Second Parsing:** Deterministic ingestion and entity extraction in under 800ms.
- **Mathematical Multi-Factor Match:** Algorithmic scoring (Skills 50%, Experience 30%, Education 20%).
- **Physical Storage Vault:** Scanned uploads linked to exact physical drawers (e.g. Cabinet B, Shelf 3).
- **AI Voice Caller:** Automated bilingual voice confirmation (Tagalog/English) reducing no-shows.

## 1.2 The Three Portal Ecosystem

The system is architected around three dedicated, role-specific web applications sharing a unified, secure backend:

### 1. Candidate Portal

Empowers applicants to search Mariwasa careers, leverage **Smart Match** (upload without selecting a job to get instant job recommendations), verify AI-extracted credentials via the **Preview & Verify** interface, track application stages in real-time, and message HR directly.

### 2. HR Screening Portal

Provides recruitment officers with high-fidelity candidate rankings, match score inspection (matched vs. missing competencies), side-by-side comparison, Google Calendar interview booking, AI Voice Caller triggers, and physical hard-copy filing management.

### 3. Admin Control Center

Gives IT and HR administrators centralized oversight over accounts and role permissions, dynamic runtime tuning of scoring weights and threshold cutoffs, rate limits, audit logging of all sensitive events, and full job requisition CRUD control.

## 1.3 High-Level Technical Architecture

ARAS employs an asynchronous, event-driven micro-service architecture optimized for high concurrency, low latency, and deterministic reliability:

| System Layer | Core Technology Stack | Architectural Responsibilities |
|---|---|---|
| **Presentation Layer** | Next.js 15 (App Router), TypeScript, Tailwind CSS v4, Redux Toolkit | Delivers responsive, server-side rendered and interactive client interfaces for candidates, recruiters, and admins with sub-100ms client route transitions. |
| **API Gateway & Services** | FastAPI (Python 3.12), Pydantic v2, SQLAlchemy 2.0 (Async) | Exposes non-blocking asynchronous REST endpoints, manages JWT bearer security, orchestrates business rules, and bridges WebSocket streams. |
| **Task Worker Engine** | Celery 5.4, Redis 7 (Alpine), spaCy 3.7 PhraseMatcher | Executes asynchronous heavy NLP parsing, background candidate match recalculations, and scheduled voice call retries without blocking HTTP threads. |
| **Data Persistence** | PostgreSQL 16 (Alpine) with `asyncpg` connection pool | Stores normalized entity models for users, candidate profiles, job descriptions, applications, interview logs, and immutable audit trails. |
| **Object Storage** | Supabase Storage (S3-compatible) | Secure cloud storage for candidate resume PDFs, profile avatars, and scanned hard-copy walk-in physical resumes. |
| **AI Voice & Gateway** | Twilio Voice API, Google Gemini 1.5/2.0 Flash, Cloudflare Tunnel | Bridges real-time bi-directional audio streams for automated candidate phone screening and provides secure external HTTPS ingress with automatic CORS. |

## 1.4 Relational Database Architecture (ERD)

The persistence layer is modeled in PostgreSQL 16 utilizing strict foreign-key referential integrity, automated cascade policies, and asynchronous connection pooling via `asyncpg`. The schema is organized into distinct logical sub-domains: User Authentication & RBAC, Candidate Data & Work History, Job Postings & Requisition Criteria, Applications & Scoring Results, Interview Scheduling & Call Logs, and Physical Resume Vault Records.

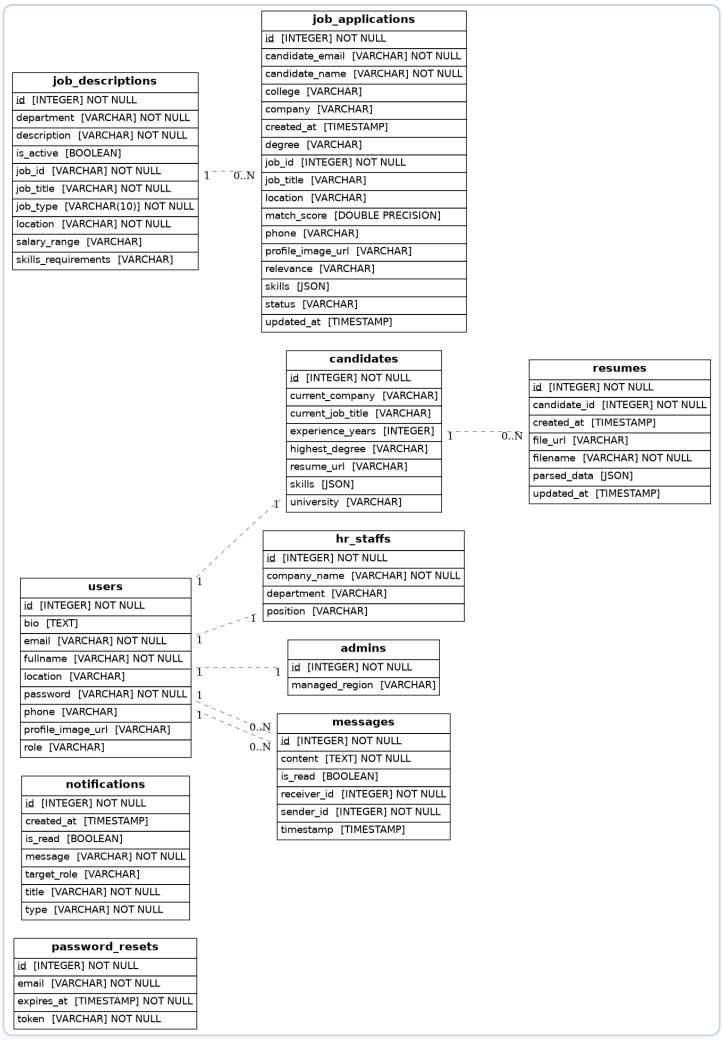



*Figure 1.2: Relational Database Schema & Entity Relationships (PostgreSQL 16)*

CHAPTER 2 • ACCESS & SECURITY

# System Access, Authentication & Role-Based Permissions

## 2.1 Access Endpoints & Network Environments

The Automated Resume Analysis System can be accessed across corporate local area networks (LAN) or securely over public HTTPS via automated Cloudflare tunnels. All HTTP traffic is enforced over TLS 1.3 encryption.

| Portal / Target Endpoint | Local Docker Environment | Production / Cloudflare Tunnel Route |
|---|---|---|
| Public Landing & Careers | `http://localhost:3000/` | `https://<subdomain>.trycloudflare.com/` |
| Candidate Dashboard | `http://localhost:3000/candidate/dashboard` | `https://<subdomain>.trycloudflare.com/candidate/dashboard` |
| HR Screening Portal | `http://localhost:3000/hr/dashboard` | `https://<subdomain>.trycloudflare.com/hr/dashboard` |
| Admin Control Center | `http://localhost:3000/admin/dashboard` | `https://<subdomain>.trycloudflare.com/admin/dashboard` |
| FastAPI Interactive Docs | `http://localhost:8000/docs` | `https://<api-subdomain>.trycloudflare.com/docs` |

## 2.2 Role-Based Access Control (RBAC) Matrix

Access to resources, actions, and data views is strictly regulated based on authenticated user roles. The system enforces zero-trust boundaries at both the frontend Next.js routing layer (via middleware and client route guards) and backend FastAPI dependency guards (`Depends(get_current_user)` and role assertions).

| System Capability / Function | Public / Guest | Candidate | HR Officer | Administrator |
|---|---|---|---|---|
| Browse Job Openings & Job Specs | YES | YES | YES | YES |
| Smart Match Resume Upload & Analysis | YES | YES | N/A | N/A |
| Submit Application & Preview-Verify | WITH ACCT | YES | NO | NO |
| Track Application Lifecycle Stages | NO | OWN ONLY | ALL | ALL |
| Candidate-HR Live Messaging | NO | OWN THREAD | ALL THREADS | ALL THREADS |
| View Candidate Scores & Missing Skills | NO | SCORE ONLY | FULL RADAR | FULL RADAR |
| Side-by-Side Candidate Comparison | NO | NO | YES | YES |
| Schedule Interviews & Google Sync | NO | RSVP ONLY | YES | YES |
| Dispatch Automated AI Voice Calls | NO | RECEIVE | TRIGGER | TRIGGER |
| Physical Hard-Copy Resume Filing | NO | NO | YES | YES |
| Create, Edit, Archive Job Requisitions | NO | NO | YES | FULL CRUD |
| User Account Provisioning & Roles | NO | NO | NO | YES |
| Configure AI Weights & Thresholds | NO | NO | NO | YES |
| View Security Audit Trail & Event Logs | NO | NO | NO | YES |

## 2.3 Registration & Authentication Procedures

1. **Candidate Self-Registration**
   New applicants navigate to `/register`. Candidates enter their Full Name, Email Address, Contact Number, and create a secure password (minimum 8 characters, requiring mixed alphanumeric characters). Upon clicking *Create Account*, the record is created in PostgreSQL with an initial role of `CANDIDATE`.

2. **HR & Admin Staff Account Provisioning**
   HR Recruiters and System Administrators cannot self-register. Staff accounts must be provisioned either through the **Admin Control Center** (`/admin/users`) or initialized via command-line database seeds (`./start.sh --seed`). This prevents unauthorized administrative privilege escalation.

3. **Sign-In & JWT Token Issuance**
   Users log in at `/login` using registered credentials. The backend verifies password hashes using `bcrypt`. Upon successful validation, the server issues a digitally signed JSON Web Token (JWT) bearing the user's ID, role, and expiration timestamp. The frontend stores this token in secure client state and injects it into every Axios HTTP request header as `Authorization: Bearer <token>`.

4. **Role-Based Automatic Redirection**
   Upon successful authentication, the system automatically redirects users to their appropriate home dashboard based on role claims:

   - `CANDIDATE` → Directed to `/candidate/dashboard`
   - `HR` → Directed to `/hr/dashboard`
   - `ADMIN` → Directed to `/admin/dashboard`

## 2.4 Password Recovery & Self-Service Reset

If a candidate or staff member forgets their password, the self-service recovery protocol ensures account access can be restored safely:

1. Navigate to `/forgot-password`.
2. Enter the registered email address and submit the request.
3. The backend generates a cryptographically secure, time-limited token stored in the `password_resets` database table (valid for 60 minutes).
4. The user receives a secure link pointing to `/reset-password?token=<token>`.
5. The user supplies a new compliant password, which is hashed with fresh salt and immediately replaces the obsolete credential.

CHAPTER 3 • CANDIDATE PORTAL

# Candidate Portal Operations Manual

## 3.1 Candidate Portal Overview & Dashboard

The Candidate Portal provides job seekers with a streamlined, transparent experience. Upon logging into `/candidate/dashboard`, applicants are greeted with a personalized command center featuring four primary summary counters:

| Total Applied | Under Review | Shortlisted | Interviews |
|---|---|---|---|
| Active submissions | Screening by HR | Qualified for next round | Scheduled sessions |

## 3.2 Finding & Filtering Job Opportunities

Candidates can browse all open requisitions at `/candidate/findjobs` (or the public `/careerspage`). The interface provides real-time search and multi-facet filtering:

- **Search Bar:** Type keywords matching job titles, required technical proficiencies (e.g. *AutoCAD*, *Lean Manufacturing*), or department names.
- **Department Filter:** Filter across *Manufacturing*, *Quality Assurance*, *Engineering & Maintenance*, *Supply Chain & Logistics*, or *Human Resources*.
- **Employment Type:** Select *Full-Time*, *Part-Time*, or *Contractual* roles.
- **Job Requisition Cards:** Each card displays the job title, department badge, plant location (Santo Tomas, Batangas), salary range, and required experience level. Clicking *View Details* opens the full job description.

## 3.3 The "Smart Match" Direct Resume Upload Workflow

A signature innovation of ARAS is the **Smart Match** capability (`/candidate/smart-upload`). Job seekers are not required to manually inspect dozens of job postings. Instead, candidates can simply upload their resume once, and the engine automatically calculates their fit across *every single active vacancy* in the organization.

1. **Drag-and-Drop Resume Ingestion**
   Upload a resume in PDF, DOCX, or plain text format (maximum file size: 10MB). The file is uploaded asynchronously to Supabase Storage.

2. **Deterministic NLP Extraction**
   The backend Celery worker and spaCy engine parse the document in under 800ms, extracting contact information, education history, work timeline, and identifying technical competencies from the curated taxonomy.

3. **Multi-Job Algorithmic Matching**
   The parsed data is matched against all open job requisitions using the Multi-Factor scoring formula (Skills 50%, Experience 30%, Education 20%).

4. **Ranked Recommendation List**
   The candidate is redirected to `/candidate/smart-matches`, displaying a ranked list of best-fitting positions ordered by match score (e.g., *Production Engineer — 92% Match*). Matched skills are highlighted with green badges, and a 1-click *Apply Now* button facilitates instant application.

## 3.4 Human-in-the-Loop "Preview & Verify" Interface

To guarantee 100% data fidelity and prevent algorithmic misinterpretation, ARAS mandates a **Preview & Verify** step prior to final job application submission (`/preview-and-verify/[jobId]`). Candidates review the raw information extracted by the AI and can edit, add, or delete any record.

| Extracted Section | Candidate Verification Actions Available | Impact on Match Score |
|---|---|---|
| **Personal & Contact** | Edit Full Name, Email, Contact Phone, and Residential Address. | Used for communication and AI Voice phone calls. |
| **Education History** | Select Degree Level (Doctorate, Master's, Bachelor's, Associate, Vocational, Senior High), Institution Name, Graduation Year. | Directly recalculates the 20% Education Hierarchy score. |
| **Work Experience** | Add/Edit employer name, job title, start/end dates, and job summary. Compute total tenure duration. | Directly recalculates the 30% Experience Relevance score. |
| **Technical & Soft Skills** | Review parsed skill tags (green pills). Add missing proficiencies or delete irrelevant terms. | Directly recalculates the 50% Skills Alignment score. |
| **Certifications** | Add licenses (e.g. PRC Licensed Engineer, TESDA NCII, Six Sigma). | Awards certification alignment bonus points. |

> **Candidate Tip for Maximum Match Scores**
> If you possess critical certifications (e.g., *PRC Mechanical Engineer* or *Lean Manufacturing*) that were abbreviated on your resume, ensure you verify them in the Preview & Verify screen before final submission. This guarantees the scoring engine credits full points.

## 3.5 Application Tracking Dashboard & Status Progression

Candidates track all submissions in real-time at `/candidate/applicationtracking`. Each application displays an interactive timeline detailing current status:

1. **SUBMITTED** INITIAL STAGE
   The application and verified resume credentials have been received and stored in the database. Awaiting recruiter review.

2. **UNDER REVIEW** IN PROGRESS
   An assigned HR recruiter has opened and reviewed the application details and verified qualifications.

3. **SHORTLISTED** QUALIFIED
   The candidate exceeded the match score threshold and has been placed on the priority hiring slate.

4. **INTERVIEW SCHEDULED** SESSION BOOKED
   An official interview session has been scheduled. The candidate receives email and calendar invitations, plus an optional automated voice notification.

5. **ACCEPTED or REJECTED** TERMINAL DECISION
   Final recruitment outcome. Candidates receive official confirmation and onboarding instructions if accepted.

## 3.6 Real-Time Candidate-HR Messaging

Candidates can communicate directly with the HR recruitment team at `/candidate/messages`. The real-time messaging interface is powered by WebSockets:

- **Inquiry Threads:** Ask questions regarding plant shift schedules, benefits, or interview locations.
- **Document Requests:** Receive requests from recruiters for supplementary credentials (e.g. TOR, NBI clearance, diploma).
- **Unread Notification Badges:** Red counter badges indicate pending messages across candidate navigation bars.

# HR Screening & Recruitment Portal Operations

## 4.1 Recruitment Dashboard & Pipeline Analytics

The HR Screening Portal ( `/hr/dashboard` ) serves as the daily operational cockpit for recruiters and hiring managers at Mariwasa Siam Ceramics Inc. The dashboard synthesizes high-level recruitment metrics into real-time KPI tiles and visual pipeline funnels:

| TOTAL APPLICANTS | PENDING REVIEW | SHORTLISTED SLATE | INTERVIEWS BOOKED |
|---|---|---|---|
| **1,428** | **184** | **92** | **36** |
| ↑ 12% vs last month | Requires HR action | Score ≥ 75% threshold | Synced with Google Cal |

Recruiters can monitor application inflow across manufacturing plant departments (Manufacturing, Quality Assurance, Maintenance, Supply Chain, and Administration) to identify pipeline bottlenecks before they disrupt production schedules.

## 4.2 Candidate Screening & Ranking Matrix

Navigating to `/hr/screening` opens the candidate evaluation grid. Unlike conventional applicant tracking systems that present unstructured chronologies, ARAS automatically ranks all applicants using the algorithmic **Multi-Factor Match Score** (0% to 100%).

### Search, Filters & Controls

- **Requisition Dropdown:** Filter candidates applied to specific job postings (e.g. *Maintenance Technician*).
- **Minimum Score Slider:** Set cutoffs (e.g., show only applicants with ≥ 70% match score).
- **Status Tabs:** View *All*, *Submitted*, *Under Review*, *Shortlisted*, or *Interview Scheduled*.
- **View Modes:** Toggle between detailed *Card View* and high-density *Table View*.

### Bulk Action Toolbar

Recruiters can select multiple candidates via checkboxes to execute batch actions in 1 click:

- **Bulk Shortlist:** Transition top candidates into the shortlisted queue.
- **Bulk Review:** Mark new submissions as Under Review.
- **Bulk Export:** Export selected candidate records to CSV or Excel.

## 4.3 Deep Candidate Inspection (Candidate Details Modal)

Clicking on any candidate card opens the comprehensive **Candidate Details Modal**. This interface provides unprecedented diagnostic clarity into the algorithmic scoring breakdown:

| Evaluation Element | Information Displayed in Inspection Modal | Recruiter Action |
|---|---|---|
| **AI Match Breakdown** | Displays the three weighted component scores: Skills Alignment (50%), Experience Relevance (30%), and Education Level (20%). | Verify scoring balance against job complexity. |
| **Matched Skills** | Highlighted in GREEN BADGES (e.g., *AutoCAD*, *Lean Six Sigma*, *Preventive Maintenance*). | Confirms required competencies are possessed. |
| **Missing Critical Skills** | Highlighted in RED BADGES. Pinpoints qualifications requested by the job description that the candidate lacks. | Determines if on-the-job plant training can bridge gaps. |
| **Experience Timeline** | Chronological employment tenure with employer names, dates, and responsibilities. | Cross-check manufacturing tenure. |
| **In-App Resume Viewer** | Integrated document previewer rendering the original PDF/DOCX file directly in the browser. | Review original document layout and formatting. |
| **Recruiter Evaluation** | Private internal text box for HR notes and 1-5 star candidate rating. | Record remarks visible only to HR staff. |

## 4.4 Side-by-Side Candidate Comparison Tool

When selecting finalists from a competitive applicant pool, recruiters utilize the **Compare Candidates Tool** ( `/hr/comparecandidates` ). This feature allows recruiters to lock 2 to 4 candidates in a comparative matrix:

1. **Select Applicants for Benchmarking**
   From the Screening Portal, select checkboxes beside 2 to 4 candidates and click *Compare Candidates*.

2. **Review Side-by-Side Competency Matrix**
   The screen renders a horizontal column for each applicant, contrasting overall AI match scores, years of relevant industry experience, highest degree earned, matched skill counts, and expected salaries.

3. **AI Comparative Fit Summary**
   The system synthesizes a comparative narrative highlighting candidate relative strengths (e.g., *Candidate A possesses superior technical certifications, whereas Candidate B brings 3 additional years of plant supervisory experience*).

## 4.5 Interview Scheduling & Google Calendar Integration

ARAS provides full-featured calendar scheduling ( `/hr/scheduling` ) with two-way Google Calendar synchronization:

- **Schedule Interview Modal:** Set interview date, time window, duration (30, 45, 60 minutes), and interview mode (*In-Person Plant Office*, *Google Meet Video Call*, or *Phone Interview*).
- **Interviewer Assignment:** Assign specific HR officers or plant department heads.
- **Google Calendar Sync:** Synchronizes scheduled events with the recruiter's official Google Calendar via OAuth 2.0. Generates a Google Meet link automatically for video sessions.
- **Candidate Invitation:** Dispatches an automated email and calendar invite ( `.ics` ) to the candidate.
- **RSVP Tracking:** Tracks whether the candidate has *Confirmed*, *Declined*, or requested a *Reschedule*.

## 4.6 Automated AI Voice Interview Caller (Twilio + Gemini Real-Time)

To eliminate candidate interview no-shows and accelerate phone screening, ARAS integrates an **Automated AI Voice Caller** powered by Twilio Voice and Google Gemini Real-Time Audio ( `/ai-caller/` ):

### Core AI Voice Capabilities

- **Automated Outbound Calling:** Trigger calls with 1 click from the interview modal ( `/ai-caller/interview/{id}/call` ).
- **Bilingual Conversational AI:** Converses fluidly in *Tagalog / Filipino* or *English*.
- **Verbal Response Interpretation:** Natural language understanding interprets spoken responses (e.g. *"Opo, makakarating po ako sa interview sa Lunes"*) or touch-tone DTMF.
- **Automated Attendance Update:** Automatically marks the interview status as *CONFIRMED* upon affirmative voice response.

### Resilience & Fallback Protocol

- **Smart Retry Policy:** If the call is unanswered, Celery schedules retries after 2 hours and 24 hours.
- **Backup SMS Notification:** Dispatches an SMS confirmation reminder if voice call attempts fail.
- **Hardware ADB Dialing:** Supports connecting a local Android smartphone via USB ADB for direct SIM dialing.
- **Call Audio & Transcripts:** Recruiter can review call duration, status, and transcripts in the interview record.

## 4.7 Physical Hard-Copy Resume Storage Vault

At the Mariwasa plant in Santo Tomas, Batangas, many local factory and maintenance workers submit physical, printed paper resumes at the guardhouse or reception desk. The **Physical Hard-Copy Resume Storage Vault** ( `/hr/physical-resumes` ) ensures walk-in submissions receive the same digital intelligence as online applications:

1. **Scan & Ingest Hard-Copy Resume**
   HR staff scan the paper resume using an office scanner or mobile camera and upload the file (PDF/JPG) at `/hr/physical-resumes`.

2. **Assign Physical Filing Location Reference**
   The recruiter logs the exact physical storage location tag (e.g. `Cabinet B - Drawer 2 - Folder 2026-Eng`). This tag is permanently bound to the digital record, allowing any HR officer to physically retrieve the original document in seconds.

3. **OCR Parsing & Requisition Linking**
   The system runs deterministic parsing on the scanned document, links the applicant to the target job requisition, and calculates the AI Match Score alongside digital applicants.

4. **Vault Search & Retrieval**
   Filter physical resumes by status (`RECEIVED`, `UNDER_REVIEW`, `SHORTLISTED`), job requisition, or physical location tag. Recruiters can click *Copy Location* to copy filing references directly to clipboard.

CHAPTER 5 • ADMIN CONTROL CENTER

# System Administration & Governance Guide

## 5.1 Admin Dashboard & Infrastructure Telemetry

The Admin Control Center (`/admin/dashboard`) gives system administrators and IT leads total visibility over operational infrastructure, active user traffic, and background task queues:

### PostgreSQL 16 Connection Pool

Monitors active and idle database connections managed by `asyncpg`. Default pool size: 10 connections with overflow ceiling of 5 and automatic pre-ping health checks.

### Redis 7 Message Broker

Tracks Celery task backlog, Celery worker ping responses, and Redis key memory utilization for the dynamic matching configuration cache.

### Cloudflare Quick Tunnel

Displays live public HTTPS tunnel ingress URL and verifies that backend CORS middleware dynamically authorizes requests from frontend tunnel endpoints.

## 5.2 User Account Management & Role Provisioning

Administrators manage all user credentials, role assignments, and security states at `/admin/users`:

| Administrative Action | Procedural Description | Security Safeguards |
|---|---|---|
| **Create Staff Account** | Click *+ Add User*. Enter staff name, email, initial temporary password, and assign role (`HR` or `ADMIN`). | Requires admin re-authentication; initial login prompts mandatory password update. |
| **Role Elevation / Demotion** | Change role dropdown between `CANDIDATE`, `HR`, and `ADMIN`. | Revokes active JWT sessions immediately; logged in audit trail. |
| **Account Suspension** | Toggle *Active / Suspended* toggle switch. | Suspended users are blocked at API middleware with HTTP 403 Forbidden. |
| **Force Password Reset** | Trigger administrative reset. Dispatches reset link to staff email. | Invalidates all existing refresh tokens. |

## 5.3 Job Requisition Management

The recruitment team relies on administrators to configure job openings at `/admin/jobmanagement`. Each job description directly influences the deterministic AI matching engine:

1. **General Job Requisition Details**
   Define Job Title (e.g. *Production Engineer*), Department (e.g. *Manufacturing*), Employment Type (Full-Time, Contract), Plant Location, and Salary Range.

2. **Technical Competency Taxonomies**
   Input comma-separated technical proficiencies (e.g., *AutoCAD, Lean Manufacturing, ISO 9001, Quality Control*). These terms are parsed by the spaCy PhraseMatcher during resume ingestion.

3. **Minimum Educational & Experience Requirements**
   Specify the required degree level (e.g., *Bachelor's degree in Engineering*) and minimum years of industry experience (e.g., *2-3 years in ceramics or manufacturing*). These parameters establish the baseline for the Education Hierarchy and Experience Relevance algorithms.

4. **Requisition Status Control**
   Toggle job states: *Active* (visible to public and eligible for Smart Match), *Draft* (internal review), or *Archived* (closed position).

## 5.4 Dynamic System Configuration & Algorithm Calibration

A critical enterprise feature of ARAS is the **Dynamic System Settings** dashboard (`/admin/system-config`). Administrators can adjust algorithmic parameters in real time without restarting backend servers or modifying source code. Settings are cached in Redis for sub-

millisecond retrieval:

| Configuration Parameter | Default Value | Adjustable Range | Algorithmic Impact & Functional Description |
|---|---|---|---|
| `skills_weight` | **0.45 – 0.50** | 0.0 – 1.0 | Relative weight of direct technical and soft skill overlap in candidate overall match score. |
| `experience_weight` | **0.25 – 0.30** | 0.0 – 1.0 | Relative weight of total career tenure and semantic job title relevance. |
| `education_weight` | **0.15 – 0.20** | 0.0 – 1.0 | Relative weight based on degree hierarchy (Doctorate > Master's > Bachelor's > Associate's). |
| `fuzzy_match_threshold` | **0.75** | 0.50 – 0.95 | Minimum string sequence similarity ratio required to consider non-exact skill terms a match. |
| `minimum_match_score` | **30.0%** | 0.0 – 80.0% | Cutoff threshold below which candidates are excluded from Smart Match job recommendations. |
| `gemini_api_keys` | **Pool** | List of Keys | Comma-separated pool of Gemini API keys rotated in round-robin fashion for qualitative LLM candidate summaries. |

## 5.5 Security Audit Logging & Compliance Trails

To adhere to corporate governance standards, ARAS records an immutable audit log for every sensitive event in the `audit_logs` database table:

- **Tracked Security Events:** User authentication attempts (success and failure), password resets, role modifications, resume downloads, manual score overrides, and job posting deletions.
- **Metadata Recorded:** User ID, Actor Email, Timestamp (UTC), IP Address, User-Agent, and detailed JSON payload describing pre-action and post-action states.
- **Audit Search & Export:** Administrators can query audit records by date range, user ID, or event category and export compliance reports for security audits.

# AI Intelligence Engine & Scoring Specifications

## 6.1 Deterministic Document Parsing Pipeline

A core architectural design tenet of ARAS is the deliberate separation between **deterministic document parsing** and **qualitative generative analysis**. By executing the primary resume ingestion deterministically, the system achieves sub-800ms parsing speeds, zero recurring LLM token expenses, and absolute mathematical consistency across all evaluations.

1. **Document Preprocessing (PyMuPDF & python-docx)**
   The parser accepts PDF, DOCX, and TXT files. PyMuPDF extracts raw text, structural bounding boxes, and typography weights, identifying section headings (e.g. *Education*, *Work Experience*, *Skills*, *Certifications*) based on font sizing, capitalization, and vertical spacing cues.

2. **Named Entity Recognition (NER) & Contact Extraction**
   Regular expression heuristics and spaCy statistical models extract candidate contact particulars: candidate full name, phone number (normalized to Philippine E.164 format: `+639XXXXXXXXX`), email address, and residential municipality.

3. **Contextual Experience & Education Timeline Segmentation**
   Date parser algorithms extract start and end dates (e.g. *"June 2021 – Present"*), calculating total months of cumulative professional experience while discounting overlapping date ranges.

## 6.2 spaCy PhraseMatcher & Technical Taxonomy

To eliminate keyword misses caused by variations in vocabulary, ARAS incorporates an enterprise taxonomy containing over **2,000+ domain terms** specifically tuned for ceramic tile manufacturing, chemical engineering, industrial automation, and corporate management:

### Exact & Lemmatized Matching

spaCy tokenizes and lemmatizes the document text. The PhraseMatcher indexes multi-token terminology (e.g. *"Root Cause Analysis"*, *"Preventive Maintenance"*, *"Kiln Operation"*) with exact token boundary verification, preventing false-positive substring matches.

### Fuzzy Sequence Matching (Threshold 0.75)

When candidates write typos or typographical variations (e.g. *"Lean 6-Sigma"* vs. *"Six Sigma"*), the difflib `SequenceMatcher` evaluates similarity against the taxonomy. If similarity exceeds `fuzzy_match_threshold` (0.75), credit is awarded.

## 6.3 Multi-Factor Match Scoring Formulation

Candidate alignment against a job requisition is computed through a standardized, multi-dimensional scoring model. The overall score represents the weighted summation of individual competency sub-scores:

ARAS ALGORITHMIC MATCH FORMULA

$$\text{Match Score} = (w_{skills} \times S_{skills}) + (w_{exp} \times S_{exp}) + (w_{edu} \times S_{edu}) + (w_{cert} \times S_{cert}) + (w_{loc} \times S_{loc})$$

Where default weights are: $w_{skills}$ = 0.45 – 0.50, $w_{exp}$ = 0.25 – 0.30, $w_{edu}$ = 0.15 – 0.20, $w_{cert}$ = 0.10, and $w_{loc}$ = 0.05. The sum of all weights equals 1.00 (100%).

### 1. Skills Alignment Sub-Score ($S_{skills}$)

Measures direct overlap between the candidate's verified skills and the mandatory competencies specified in the job posting:

```
S_skills = min(100.0, (len(matched_required_skills) / len(total_required_skills)) * 100.0 + bonus_preferred)
```

### 2. Experience Relevance Sub-Score ($S_{exp}$)

Evaluates cumulative verifiable professional experience relative to the minimum years demanded by the requisition:

```
if candidate_years >= required_years:
    S_exp = 100.0
else:
    S_exp = (candidate_years / required_years) * 100.0
```

### 3. Education Hierarchy Sub-Score ($S_{edu}$)

Academic credentials are mapped to a standardized degree hierarchy scale from 1 (Senior High) to 6 (Doctorate / PhD):

| Hierarchy Level | Academic Qualification Classification | Base Score Assigned | Relevance to Mariwasa Positions |
|---|---|---|---|
| **Level 6** | Doctorate / Ph.D. / Post-Graduate | 100 PTS | Advanced research, materials science, executive roles. |
| **Level 5** | Master's Degree (MBA, M.S. Engineering) | 95 PTS | Senior plant management, specialized engineering. |
| **Level 4** | Bachelor's Degree (B.S. Chem/Mech/IE, etc.) | 80 PTS | Standard requirement for production engineers, QA supervisors. |
| **Level 3** | Associate's Degree / 3-Year Technical Course | 60 PTS | Laboratory technicians, junior line supervisors. |
| **Level 2** | Diploma / Vocational Course / TESDA NCII | 40 PTS | Standard requirement for maintenance technicians, operators. |
| **Level 1** | Senior High School (SHS) Graduate | 20 PTS | Entry-level plant packaging, warehousing personnel. |

## 6.4 Gemini LLM Integration & Hybrid Fallback Architecture

For shortlisted candidates, ARAS invokes Google Gemini 1.5/2.0 Flash to synthesize a qualitative narrative summary and identify subtle contextual nuances. To ensure enterprise-grade resilience against third-party API rate limits, the system operates a **Hybrid Fallback Architecture**:

- **Rotational API Key Pool:** The system maintains multiple Gemini API keys in `GEMINI_API_KEYS`, cycling through them in a round-robin rotation to avoid per-minute quota ceilings.
- **Deterministic Fallback:** If all external API keys fail or internet access is severed, the backend automatically falls back to 100% deterministic rule-based scoring without throwing exceptions or delaying user workflows.

## 6.5 Twilio + Gemini Real-Time Voice Streaming Architecture

The AI Voice Caller establishes a bi-directional audio bridge between Twilio's telephony infrastructure and Gemini's Real-Time speech models via WebSockets ( `/ai-caller/media-stream` ). Audio is sampled at 8kHz mu-law telephony standards, converted to PCM, and processed in real time to interpret candidate verbal commitments in Tagalog and English with sub-400ms latency.

CHAPTER 7 • OPERATIONS & DEPLOYMENT

# Deployment, CLI Operations & Environment Setup

## 7.1 Container Architecture & Service Topology

ARAS is packaged into isolated, production-ready Docker containers orchestrated via `docker-compose.yml`. The container topology isolates compute, storage, and worker queues to maximize reliability and horizontal scalability:

| Container Service | Base Image | Internal Port | Role & Operational Function |
|---|---|---|---|
| **frontend** | Node 20 / Next.js 15 | `3000` | Next.js App Router frontend server rendering UI and serving static assets. |
| **backend** | Python 3.12 (Slim) | `8000` | FastAPI application handling HTTP API requests, JWT auth, and WebSocket streams. |
| **celery_worker** | Python 3.12 (Slim) | `N/A` | Asynchronous Celery worker executing heavy spaCy parsing and background match scoring. |
| **redis** | Redis 7.2 (Alpine) | `6379` | In-memory broker for Celery queues and low-latency cache for matching weights. |
| **db** | PostgreSQL 16 (Alpine) | `5432` | Relational database persisting user accounts, job postings, applications, and logs. |
| **cloudflared** | Cloudflare Quick Tunnel | `N/A` | Establishes encrypted public HTTPS ingress tunnels with automatic CORS configuration. |

## 7.2 The Unified Management CLI ( `./start.sh` )

ARAS provides an all-in-one automation script ( `./start.sh` ) located at the root of the repository. This script handles Docker container orchestration, Cloudflare tunnel generation, environment synchronization, and health verifications.

| CLI Command Syntax | Functional Description & Use Case |
|---|---|
| `./start.sh` | Standard start. Builds images, launches all containers, spins up a live Cloudflare tunnel, and synchronizes CORS settings. |
| `./start.sh -q` or `--quick` | Fast start. Launches containers without checking or rebuilding Docker image layers. Ideal for daily development. |
| `./start.sh -b` or `--rebuild` | Force clean rebuild. Rebuilds all containers with `--no-cache` before launching services. |
| `./start.sh -r` or `--restart` | Restart stack. Gracefully cycles all containers and regenerates fresh Cloudflare tunnel URLs. |
| `./start.sh -s` or `--stop` | Graceful shutdown. Stops all running application containers and terminates active background tunnels. |
| `./start.sh --status` | Health check. Inspects container runtime statuses and verifies live Cloudflare tunnel health. |
| `./start.sh --logs [service]` | Log streamer. Streams live aggregated logs or targets a specific service (e.g. `./start.sh --logs backend` ). |
| `./start.sh --seed` | Database seeder. Populates default Mariwasa job openings and default HR staff accounts. |

## 7.3 Environment Configuration ( `.env` ) Parameter Glossary

The central configuration file ( `.env` ) governs system security, database connections, and third-party integrations. All sensitive secrets must be configured prior to production deployment:

| Environment Variable | Example / Configuration Format | Operational Description |
|---|---|---|
| `DATABASE_URL` | `postgresql+asyncpg://user:pass@db:5432/ar_db` | Asynchronous PostgreSQL connection URI utilized by SQLAlchemy and `asyncpg`. |
| `REDIS_URL` | `redis://redis:6379/0` | Redis broker connection string for Celery background tasks and system cache. |
| `SECRET_KEY` | `32+ character cryptographic secret string` | Cryptographic secret used for signing and verifying JWT authentication tokens. |
| `FRONTEND_URL` | `http://localhost:3000,https://*.trycloudflare.com` | Comma-separated list of authorized origins permitted through FastAPI CORS middleware. |
| `GEMINI_API_KEYS` | `key_1,key_2,key_3` | Comma-separated pool of Google Gemini API keys rotated in round-robin fashion. |
| `GOOGLE_CLIENT_ID` | `<oauth-client-id>.apps.googleusercontent.com` | Google OAuth 2.0 client ID for Google Calendar two-way interview synchronization. |
| `GOOGLE_CLIENT_SECRET` | `<oauth-client-secret>` | Google OAuth 2.0 client secret for calendar token exchange. |
| `SUPABASE_URL` | `https://<project>.supabase.co` | Supabase Storage API URL for persisting resumes, scans, and avatars. |
| `SUPABASE_SERVICE_ROLE_KEY` | `eyJhbGciOi... (Service Role JWT)` | Privileged Supabase service role key for automated bucket uploads. |
| `TWILIO_ACCOUNT_SID` | `ACXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX` | Twilio account SID for automated voice telephone dispatch. |
| `TWILIO_AUTH_TOKEN` | `<twilio-auth-token>` | Twilio authentication token for API authorization. |
| `TWILIO_PHONE_NUMBER` | `+1XXXXXXXXXX` | Registered Twilio phone number used as caller ID for outbound interview calls. |

## 7.4 Database Reset & Migration Protocol

When deploying to a clean staging environment or wiping test data, administrators execute `backend/reset-db.sh`:

1. Drops and recreates the PostgreSQL `public` schema, purging all obsolete tables.
2. Wipes temporary storage in `uploads/resumes/` and `uploads/resume_images/`.
3. Executes `init-db.py` to initialize all SQLAlchemy ORM tables with foreign key constraints.
4. Executes `./start.sh --seed` to insert Mariwasa job requisitions (Production Engineer, QA Supervisor, Maintenance Tech) and default HR credentials.

CHAPTER 8 • TROUBLESHOOTING & REFERENCE

# Troubleshooting, FAQs & System Appendices

## 8.1 Diagnostics & Operational Recovery Procedures

This section documents structured recovery procedures for the most common technical anomalies encountered during production operation:

| Diagnostic Symptom | Probable Root Cause | Standard Resolution Procedure |
|---|---|---|
| **Resume upload stuck on "Parsing..."** | Celery worker is offline or Redis broker connection timed out. | Run `./start.sh --status` to verify worker health. If stalled, execute `docker-compose restart celery_worker redis`. Check logs via `./start.sh --logs celery_worker`. |
| **Frontend reports CORS / Network Error** | Cloudflare tunnel regenerated a new URL that is not yet reflected in backend CORS origins. | Execute `./start.sh -r` to cleanly restart all containers and automatically synchronize new tunnel URLs into backend CORS middleware. |
| **Resume upload returns HTTP 403 Forbidden** | Invalid or expired `SUPABASE_SERVICE_ROLE_KEY` in `.env`. | Verify Supabase credentials in `.env`. Confirm that storage buckets `resumes`, `profile_images`, and `resume_images` exist and have appropriate access policies. |
| **LLM Summary displays "Score calculated via deterministic rules"** | Gemini API keys exceeded daily quota or encountered internet latency. | This is normal system resilience behavior. The deterministic scoring engine operates unaffected. To restore LLM summaries, add fresh keys to `GEMINI_API_KEYS` in `.env` or via Admin Settings. |
| **Automated Voice Call fails to connect** | Candidate phone number is not in E.164 format, or Twilio account balance is depleted. | Verify candidate phone number begins with `+639...`. In development environments, utilize the **Simulate Voice Call** feature on the interview details modal to test flow without Twilio telephony charges. |

## 8.2 Frequently Asked Questions (FAQ)

### Candidate FAQs

**Q: What file formats are supported for resume upload?**
A: The system natively supports PDF (`.pdf`), Microsoft Word (`.docx`), and plain text (`.txt`) files up to 10MB in size.

**Q: Can I fix an error if the AI misread my degree?**
A: Yes! The *Preview & Verify* screen specifically allows you to edit, add, or delete any extracted degree, skill, or job experience before you submit.

**Q: What does my Match Score mean?**
A: It indicates how closely your verified qualifications match the specific job's technical competencies, experience level, and educational criteria.

### HR Recruiter FAQs

**Q: How are applicants ranked on the screening portal?**
A: Applicants are automatically sorted by their overall AI Match Score (highest to lowest), factoring in skills (50%), experience (30%), and education (20%).

**Q: How do I file walk-in printed resumes?**
A: Scan the paper resume, upload it at `/hr/physical-resumes`, and enter the physical filing reference (e.g. *Cabinet B, Drawer 1*). The AI will parse the scan automatically.

**Q: Does the AI Voice Caller replace HR interviews?**
A: No. The AI Caller conducts automated notification and preliminary attendance confirmation calls to reduce interview no-show rates.

## 8.3 Application Status Lifecycle State Machine

All job applications transition through well-defined lifecycle states governed by strict validation rules:

| Status Identifier | Portal Display Badge | State Semantics & Allowed Next Transitions |
|---|---|---|
| `SUBMITTED` | SUBMITTED | Initial submission stage. Candidate data parsed. Can transition to `UNDER_REVIEW` or `REJECTED`. |
| `UNDER_REVIEW` | UNDER REVIEW | Recruiter is inspecting profile. Can transition to `SHORTLISTED` or `REJECTED`. |
| `SHORTLISTED` | SHORTLISTED | Applicant qualified for interview round. Can transition to `INTERVIEW_SCHEDULED` or `REJECTED`. |
| `INTERVIEW_SCHEDULED` | INTERVIEW BOOKED | Official interview time booked. Can transition to `ACCEPTED` or `REJECTED`. |
| `ACCEPTED` | ACCEPTED | Terminal state. Candidate received formal job offer and onboarding package. |
| `REJECTED` | REJECTED | Terminal state. Candidate qualifications did not meet criteria for this requisition. |

## 8.4 System Specifications & Supported Browsers

ARAS is engineered for cross-platform compatibility across modern desktop and mobile browsers:

- **Google Chrome / Chromium:** Version 100+ (Full desktop & mobile support recommended).
- **Mozilla Firefox:** Version 110+ (Full support).
- **Apple Safari:** Version 16+ (macOS & iOS Safari supported).
- **Microsoft Edge:** Version 100+ (Chromium-based full support).

## 8.5 Support & Corporate Directory

| Department / Role | Contact Channel | Operational Responsibility |
|---|---|---|
| **Mariwasa HR Operations** | `hr.recruitment@mariwasa.com` | Candidate inquiries, interview scheduling, and hiring workflows. |
| **IT Operations Helpdesk** | `it.support@mariwasa.com` | User account provisioning, password lockouts, and system access. |
| **ARAS Systems Engineering** | `aras-dev@mariwasa.com` | Server deployment, container maintenance, and scoring algorithm calibration. |

> **Official Document Authentication**
>
> This document represents the authorized enterprise operational specification for the **Automated Resume Analysis System** at **Mariwasa Siam Ceramics Inc.** Version 2.4.0. Effective November 2026.